# Generative Artificial Intelligence (AI) in K-12 Classrooms Guidance

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OREGON DEPARTMENT OF EDUCATION

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## Getting Started: How to Use this Guidance

This document is grounded in a human-centered, equity-focused approach, emphasizing that the use of genAI should strengthen, not replace, authentic classroom relationships, professional experience, and meaningful student learning. It is not intended to dictate a single pathway for implementing genAI. Instead, it offers a shared framework for understanding, planning, and evaluating the use of genAI, recognizing that local contexts, community priorities, and individual student needs will shape decisions across districts and schools.

**The guidance is organized into three parts:**

- **PART 1:** Framing Responsible and Human-Centered GenAI Use - establishes the values, principles, and long-term considerations that should guide decisions.
- **PART 2:** Building GenAI Capacity, Literacy, and Safeguards in Your School - focuses on what educators and leaders can do now to prepare for responsible and ethical implementation.
- **PART 3:** Navigating Challenges and Opportunities in GenAI Implementation - addresses key issues such as equity, privacy, student health, academic integrity, and instructional practices.

Educators are encouraged to approach this guidance as a flexible resource, using the sections most relevant to their role, while maintaining a shared understanding of the essential foundational principles outlined in Part 1.

**District Leaders**

District leaders are encouraged to approach GenAI as a system-level leadership and planning priority, establishing clear policies and guardrails before classroom use and coordinating across instructional, technology, and student support teams. Their role includes aligning implementation with learning goals, equity priorities, and legal requirements, while building staff capacity and engaging families and communities in transparent decision-making.

**Recommended Sections:**

- 1a. Guiding Principles for Responsible and Ethical Use of GenAI in Education
- 1b. Keeping Humans In The Driver's Seat
- 1c. Long-term Impacts of AI On Our Students' Futures
- 2a. Building Capacity: Core AI Competencies for Educators
- 2f. Create Policy Guardrails Before GenAI Enters Your District's Classrooms
- 2g. Understanding Agentic AI and Its Implications for Schools
- 3a. Understanding Equity Issues Raised by GenAI in K-12 Settings
- 3c. Ensuring Staff and Student Privacy in School Settings
- 3e. GenAI Copyright and Licensing Guidance for School Leaders

**School Building Leaders**

Building leaders are responsible for translating district direction into coherent, school-wide practice. This includes supporting educator capacity, setting clear expectations for classroom use, and ensuring that GenAI supports durable skill development and strong instruction. They also help establish consistent approaches to academic integrity and monitor the impact of GenAI on teaching and learning.

**Recommended Sections:**

- 1a. Guiding Principles for Responsible and Ethical Use of GenAI in Education
- 1b. Keeping Humans In The Driver's Seat
- 2a. Building Capacity: Core AI Competencies for Educators
- 2b. Deeper Understanding of Durable Skills
- 2c. Information and AI Literacy as Essential Durable Skills
- 2d. The Risks of Our Students Depending Too Much on GenAI Instead of Their Own Thinking
- 3a. Understanding Equity Issues Raised by GenAI in K-12 Settings
- 3b. How GenAI Tools Can Help Support All Students
- 3d. Addressing Plagiarism and Other Integrity Concerns in the Age of GenAI

**Classroom Educators**

Teachers working directly with students are central to ensuring that GenAI is used in meaningful and developmentally appropriate ways. This includes making intentional decisions about when to use GenAI to support learning and when to prioritize independent thinking, designing instruction that emphasizes critical thinking and creativity, and teaching students to use AI tools safely, ethically, and responsibly while protecting student privacy.

**Recommended Sections:**

- 1a. Guiding Principles for Responsible and Ethical Use of GenAI in Education
- 1b. Keeping Humans In The Driver's Seat
- 2a. Building Capacity: Core AI Competencies for Educators
- 2c. Information and AI Literacy as Essential Durable Skills
- 2d. The Risks of Our Students Depending Too Much on GenAI Instead of Their Own Thinking
- 3a. Understanding Equity Issues Raised by GenAI in K-12 Settings
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Across all roles, this guidance reinforces a consistent principle: **GenAI should be used to extend human thinking, not replace it.** By grounding decisions in shared values, intentionally building capacity, and proactively addressing risks, Oregon educators and leaders can approach generative AI in ways that support student learning, well-being, and future

readiness while maintaining the essential human relationships and expertise at the core of education.

## GenAI in K-12 Classrooms

As artificial intelligence becomes an increasingly familiar presence in education and in our daily lives, the focus is shifting beyond simply defining what AI is. It is moving toward thoughtfully determining why, when, and whether it should be used at all. When its use is deemed appropriate, the emphasis then turns to how AI can be integrated safely and effectively into schools. This progression underscores the importance of prioritizing student skill and knowledge development while leveraging AI in ways that strengthen personal connections, enhance teaching and learning, expand opportunities for all students, and prepare them for an ever-changing workforce.

While AI typically refers to computer systems designed to perform tasks that require human intelligence, generative AI creates new content, including text, images, video, or computer code, based on patterns from vast datasets. This guidance primarily focuses on genAI, given its rapid emergence in schools, but references AI more generally where appropriate. It shifts from introducing genAI to educators toward practical guidance on safe, effective implementation, with the goal of enhancing student outcomes and preparing them for the future.

As AI tools become more common across instruction and operations, Oregon school districts will find it helpful to approach AI use as a system-level leadership and planning consideration rather than only an individual classroom choice. Because these tools can influence teaching practice, student learning, data privacy, and community trust, districts should consider establishing clear local policies and processes for reviewing and guiding how AI is potentially introduced and used. Coordination across instructional leadership, technology, and student support teams helps ensure new tools align with learning goals, support educator expertise, and are regularly reviewed for impact on students and staff. Communicating transparently with the school community about if, how, and why AI tools may be used can strengthen understanding and trust.

Families are central to how children use AI, yet many parents and caregivers feel unprepared. A 2026 global study from the Brookings titled, A New Direction for Students in an AI World: Prosper, Prepare, Protect, found that parents have limited awareness of AI's risks and benefits, but the majority want support in understanding and guiding their children's use. Schools cannot manage AI in isolation. Partnering with families and the broader school community to establish consistent expectations at home and school will be essential as AI becomes increasingly embedded in students' daily lives.1

While this guidance is intended primarily for school leaders and classroom educators, parents and caregivers can use it to learn more about ODE's recommendations for the safe, effective implementation of genAI, while enhancing student outcomes and preparing them with the durable skills they will need. Parents and caregivers can utilize ready-to-use family resources, such as Common Sense Media's AI Literacy Toolkit for Families, to help start conversations with their children around their use of technology, including genAI.

## PART 1: Framing Responsible and Human-Centered Generative AI (GenAI) Use

- **1a. Guiding Principles for Responsible and Ethical Use of GenAI in Education**
- **1b. Keeping Humans In The Driver's Seat**
- **1c. Long-term Impacts of AI On Our Students' Futures**

### 1a. Guiding Principles for Responsible and Ethical Use of GenAI in Education

As generative AI continues to reshape education, it is crucial for school leaders and educators to have a clear, values-driven framework to help guide decisions about **if, when, and how** to integrate this powerful technology. ODE's **AI Guiding Principles** center on responsible and equitable use to empower educators and students, supported by foundational principles that ensure these tools help enrich learning while maintaining essential human connections. These principles prioritize ethical practice and future readiness while promoting transparency, inclusivity, and continuous evaluation to prepare students as creators, critical thinkers, and problem solvers in an AI-driven world. At the heart of this vision is a pivotal question:

**How can we use generative AI to improve student learning and well-being today, while preparing them with the creative and essential skills they'll need to tackle the challenges of tomorrow?**

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# Guiding Principles for GenAI in K–12 Education

**We must empower our young people to stand as ethical creators, not passive consumers, of the technologies that will define their time. Through education that sharpens their thinking, strengthens their literacy, and grounds them in responsibility, we prepare students to shape a future built on integrity and human ingenuity.**

## I. Human-Centered Learning Comes First

AI must never replace authentic relationships between educators, students, and peers. Its use should protect and strengthen students' critical thinking, creativity, and independence, not diminish them. The use of AI should also support, not replace, the professional expertise and judgment of licensed educators, whose understanding of students, context, and learning needs remains essential to instructional decision-making.

**Guiding Question:** How will genAI be used to support, not replace, student engagement, well-being, critical thinking, and teacher-student or peer-to-peer relationships?

## II. Purposeful and Research-Based Adoption

The integration of AI is most effective when it can demonstrably enhance learning, engagement, and instructional quality. If adoption is to take place, use should be transparent, safe, and grounded in sound pedagogical reasoning. Decisions about whether and how to use genAI should remain guided by educator expertise, local instructional priorities, and established District policy.

**Guiding Question:** What criteria and evidence will we use to evaluate the accuracy, reliability, safety, and educational value of genAI tools, and are *non-AI alternatives sufficient to meet the need?*

## III. Equity and Access for All Learners

Every student benefits from having equitable access to the same technologies and tools, instruction, and literacy, ensuring all students are prepared to thrive and lead in an increasingly AI-driven world.

**Guiding Questions:**

- How will we ensure that all students, including those with disabilities, multilingual learners, and those furthest from opportunity, can benefit from safe, meaningful AI use?

- How will we address disparities in access to devices, connectivity, and the foundational digital literacy needed to engage with genAI?

## IV. Ethical Use and Informed Practice

Districts that implement AI effectively establish clear policies for safe and ethical use and provide sustained professional learning for educators, staff, and students. Instruction should likely address genAI bias, disinformation, data privacy thinking, and academic integrity to foster critical and ethical engagement.

**Guiding Questions:**

- What ongoing professional development, student learning opportunities, and policy safeguards must be in place to ensure ethical, safe, and transparent genAI use?
- How will students be taught to identify genAI bias, avoid misinformation, and uphold academic integrity?

## V. Continuous Evaluation, Future Readiness, and Community Partnership

The use of AI in education is most effective when it is regularly assessed to ensure it advances learning outcomes, centers equitable practices, and addresses sustainability and safety. Schools play an important role in preparing students for the world they are entering and not just the one they inhabit today. Ongoing accountability and inclusive engagement with the whole school staff, students, families, and the school community are essential.

**Guiding Questions:**

- Who is responsible for monitoring AI use and ensuring it aligns with up-to-date research, equity goals, and student safety?
- How will school leaders regularly evaluate AI's impact on teaching, learning, and future readiness across student groups?
- How will school staff, students, families, and the larger school community be included as partners in the evaluation of AI?

## VI. Future-Ready Learning and Workforce Preparation

AI integration can be beneficial if it supports students in developing the durable skills, knowledge, adaptability, and ethical grounding they need to succeed in the careers and civic life of the future.

**Guiding Questions:**

- How will our AI policy and implementation strategies ensure all students have the opportunity to develop the critical, creative, technical, and durable skills to not just use, but design and shape the technologies of the future?

- How are our AI-implementation policies and strategies helping to ensure that students use genAI as a tool to extend, rather than replace, their own critical thinking, creativity, and memory to help prevent cognitive offloading or atrophy of essential skills?

### 1b. Keeping Humans in The Driver's Seat

AI tools designed for schools are rapidly expanding, and according to a Clackamas ESD article, *A Sold-out AI Empowered EDU 2025 Brings Together Educators from Across the Region*, more Oregon districts are adopting them for daily use. See Key Terms Important for Understanding Artificial Intelligence for key terms for this rapidly expanding technology. While AI has long been part of educational technology, stronger evidence is still needed to understand its full impact. *Guidance from the U.S. Department of Education's Office of Educational Technology* encourages keeping humans in control, emphasizing that teachers and learners must retain the agency to interpret information and make decisions:

*We envision a technology-enhanced future more like an electric bike and less like robot vacuums. On an electric bike, the human is fully aware and fully in control, but their burden is less, and their effort is multiplied by a complementary technological enhancement. Robot vacuums do their job, freeing the human from involvement or oversight.*

**It remains essential that:**

- **Educator knowledge and expertise are honored in decision-making when implementing genAI in schools.**
- **Ongoing high-quality training and education on the responsible, ethical, and productive use of generative AI tools is provided to students and staff.**
- **Any AI tool approved for use by students has been vetted by district staff and is compliant with federal and state data privacy laws.**
- **Students build strong content knowledge and skills so they can understand the world around them and critically understand, evaluate, and responsibly use AI outputs.**

### 1c. Long-term Impacts of AI On Our Students' Futures

As genAI continues to rapidly reshape education, its rise in popularity raises big questions about the future our students are stepping into. What jobs will exist after graduation? How will AI impact our environment? Will these tools narrow opportunity gaps or widen them? While AI can expand access to learning, its use also risks deepening inequalities without strong, well-developed human and equity-centered policies and practices. With genAI now regularly embedded into digital learning platforms and helping to make curricular decisions, such as generating complete lesson plans for classroom educators according to the NEA article, *Teaching and Learning in the Age of Artificial Intelligence*, school leaders are recommended to act thoughtfully and proactively.

The authors of the recent publication, *Time for a Pause: Without Effective Public Oversight, AI in Schools Will Do More Harm Than Good*, from the National Education Policy Center at the University of Colorado Boulder, School of Education, put it succinctly:

> *Yet, important discussions about AI's potentially negative impacts on education are being overwhelmed by relentless rhetoric promoting its alleged ability to positively transform teaching and learning. The result is that AI, with little public oversight, is on the verge of becoming a routine presence in schools.*

The NEPC report urges school leaders to slow down the process, take time to ask critical questions, and ensure adoption happens responsibly, ethically, and productively. It also emphasizes that any implementation of these tools will benefit from being paired with ongoing AI training, specifically including AI literacy training, for all school staff and students.

This caution fits a broader emerging pattern. In 2026 Congressional written testimony, neuroscientist Dr. Jared Cooney Horvath shared that international test data consistently show more classroom screen time linked to *lower scores in reading, math, and science*. A growing body of research is starting to show that educational technology and digital tools (save adaptive practice programs) are not substitutes that outperform regular classroom teaching. This track record suggests schools should likely be realistic about what technology can and can't do.

ODE's Developing Policy and Protocols for the Use of Generative AI in K-12 Classrooms guidance document is a step-by-step guide for Oregon school leaders navigating this uncertain landscape, helping to ensure decisions reflect today's realities while preparing students for an AI-shaped future.

## Three Important Issues for Educational Leaders to Consider:

**1. A Fast-Changing Future Workforce for Our Students:** AI is rapidly transforming our world and our global workforce in ways that challenge longstanding trajectories for education, employment, and economic mobility.

Advances in artificial intelligence, particularly generative AI, are reshaping industries, streamlining long-standing operations, and eliminating many entry- and mid-level jobs that once served as stepping stones for young workers according to a 2025 article titled, *Canaries in the Coal Mine? Six Facts about the Recent Employment Effects of Artificial Intelligence*. A recent August 2025 study from Stanford researchers stated that, "...large-scale evidence consistent with the hypothesis that the AI revolution is beginning to have a significant and disproportionate impact on entry-level workers in the American labor market."[11] This is true for many "less exposed" industries beyond tech-specific fields. As the leaders who help shape and cultivate the future trajectories of our students, we should be thinking about what ramifications these changes are quickly going to have on our students, particularly students who have historically been and are currently being under-resourced and underserved.

In May 2025, Anthropic CEO Dario Amodei was quoted in the Axios article, *Behind the Curtain: A white-collar bloodbath* interview saying, "AI could wipe out half of all entry-level white-collar jobs and spike unemployment to 10-20% in the next one to five years."[12] In January of 2025, Meta CEO Mark Zuckerberg said in a podcast interview that, "Probably in 2025, we at Meta, as well as the other companies that are basically working on this, are going to have an AI that can effectively be a sort of mid-level engineer that you have at your company that can write code."[13]According to a May 2026 Challenger Report, *Volatility Continues: Job Cuts Up 38%, AI Leads Reasons for Second Consecutive Month*, technology companies continue to announce large-scale cuts, they are "often citing AI spend and innovation… Regardless of whether individual jobs are being replaced by AI, the money for those roles is." Job loss across industries such as banking, finance, tech, and other historical white-collar sectors is already seeing the beginnings of layoffs, with AI being cited as the main cost-cutting reason. Workers themselves, across workforce sectors, are increasingly worried that AI will reduce jobs in the industries in which they are specifically employed.[14]

While none of this is predetermined, it is something for all educational leaders to understand. From major tech firms to financial institutions and customer service sectors, employers are rapidly integrating AI into daily operations. These shifts are no longer hypothetical. Corporate layoffs tied to automation and AI adoption are now regular headlines, and even fields once considered safe (e.g., computer science and computer engineering) are seeing rising unemployment rates among recent college graduates,

according to the CBS News article, *Recent College Graduates Face a New Obstacle in Finding a Job: AI*.

**2. Transitioning Learning Goals for Our Students:** Adapting to this new reality will affect not only the use of classroom technology and instructional practices, but also the skills students truly need to thrive in a world where **durable skills**, including adaptability, creativity, critical thinking, and information and AI literacy, matter more than rote learning and memorization according to the NCSL *An AI-Ready Workforce Will Need 'Durable Skills'* article. Ignoring this shift risks deepening opportunity gaps and leaving students, especially those from our underserved communities, unprepared for a very different future than the one schools were originally designed to serve. Career readiness now includes the ability to work alongside AI, not just compete with it, and to look at career pathways where technology (e.g., working with computers as the main form of worker input) is not always a core component of day-to-day work.

This transition calls for urgent and proactive action. Guidance, curriculum, and instruction may need to be examined and possibly revised to better prepare students for an economy reshaped by AI, including its environmental footprint and influence on labor markets. Efforts to prepare students to thrive in the future workforce will benefit from bold thinking, cross-sector collaboration, and a sustained commitment to equity and access across the K-12 system.

As genAI accelerates change across every sector, education needs to move beyond simply placing new tools in classrooms. Durable skills (discussed in more detail in section 2b below) like problem-solving, critical thinking, communication, empathy, adaptability, and AI literacy should be emphasized as the backbone of daily instruction. These are the skills that enable students to adjust to rapid career disruption, collaborate effectively in diverse settings, and engage with technology thoughtfully. By embedding durable skills into core learning, schools can help ensure students are not only workforce-ready but also positioned to lead and contribute meaningfully in a society transformed by AI.

**3. AI, Greenhouse Gas Emissions, and Water Use:** Digital technologies, including AI, media streaming, cloud computing, and video conferencing, can have significant environmental impacts, including greenhouse gas (GHG) emissions and water use. These impacts are linked to the data centers that power many modern digital tools, which require large amounts of electricity and water for computing and cooling. As the use of digital technologies continues to grow, as detailed in the MIT News article, *Explained: Generative AI's Environmental Impact*, the demands of shared infrastructure place increasing pressure[15] on energy systems and local water resources.

AI is one part of a broader digital ecosystem and has received increased attention because training and operating its advanced models can be very resource intensive. At the same time, AI technologies are now being used to improve sustainability in areas such as agriculture,[16] energy efficiency, and methane leak detection, with the potential to reduce emissions and resource use.

Public disclosures highlight the scale of these issues. In Google's 2025 environmental report[17], the company reported approximately 11.5 million metric tons of GHG emissions in 2024, about a 51% increase compared to its 2019 baseline, with overall emissions driven by expanding data center infrastructure, electricity demand, and supply chain impacts tied to AI and cloud services. Water use projections raise similar concerns. Reporting from the Food & Environment Reporting Network[18] shows that new data centers built in eastern Oregon are drawing on already strained aquifers, with facilities pulling tens of millions of gallons of groundwater each year, intensifying pressure on drinking water supplies and compounding an existing pollution crisis in agricultural regions.

It is important to understand both the positive and negative impacts of using modern digital tools in context. As the *New Data: AI Is Almost Green Compared To Netflix, Zoom, YouTube* article from Forbes explains, common digital activities, such as extended media streaming or frequent video conferencing, can also consume significant resources and, depending on the amount of use, generate carbon emissions comparable to those of common genAI tasks.[19] Recognizing this broader context helps educators develop awareness and a more balanced understanding of these environmental impacts, and reinforces the idea that decision-making about the potential use of digital tools could include considerations of the environmental impacts of all digital technologies, not just AI.

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## PART 2. Building GenAI Capacity, Literacy, And Safeguards In Your School

**2a.** Building Capacity: Core AI Competencies for Educators

**2b**. Deeper Understanding of Durable Skills

**2c.** Information and AI Literacy as Essential Durable Skills

**2d.** The Risks of Our Students Depending Too Much on GenAI Instead of Their Own Thinking

**2e.** Concerns About Forming Human-Like Relationships with AI

**2f.** Create Policy Guardrails Before GenAI Enters Your District's Classrooms

**2g.** Understanding Agentic AI and Its Implications for Schools

Understanding the long-term impacts of AI, including its reshaping of the workforce, the urgency of emphasizing durable skills, and its environmental footprint, highlights the scale of change facing today's students.

While these forces will continue to evolve, educational leaders can work now to help shape how schools respond. The next section turns from what is coming to what can be done now, focusing on deepening knowledge of durable skills, strengthening digital and AI literacy, addressing the risks of cognitive offloading, and establishing thoughtful policy guardrails before genAI becomes embedded in daily classroom practice.

### 2a. Building Capacity: Core AI Competencies for Educators

Building capacity is not a one-time event; it is a systematic approach to ensuring educators feel confident, ethical, and effective when using genAI. To support this, school and district leaders could focus on developing five "Required Competencies" among all instructional staff:

- **Operational Knowledge:** Educators should consider learning about the basic mechanics of LLMs, specifically that they are predicting the next word rather than searching a database of facts. This foundational knowledge helps staff understand why errors can occur and why content generated by AI requires human verification.
- **Prompt Engineering:** Effective use of genAI requires the ability to "drive" the machine. Educators should be competent in crafting clear, context-rich prompts that include a specific persona, task, audience, and format. Educators should be able to iterate, refining their prompts based on AI output to generate high-quality, accurate content.
- **Critical Evaluation:** Because genAI can mirror societal biases and inequities present in available data, educators must be able to critically evaluate genAI output and content.

To ensure the use of high-quality instructional materials reflective of an equity-centered classroom, educators should have the ability to double-check AI-generated content for possible stereotypes, cultural inaccuracies, and a lack of diverse perspectives and backgrounds.

- **Instructional Integration:** Beyond administrative efficiency, educators need the competency to weave AI into the learning process. This includes knowing when to allow AI and when to restrict it to protect the "productive struggle" of learning. Educators should be able to redesign assessments to be "AI-resistant" or "AI-integrated," moving away from simple recall toward higher-order thinking.
- **Data Privacy and Policy:** Educators must be competent in the ethical "rules of the road," including awareness of student data privacy laws (FERPA/COPPA) and their district's Acceptable Use Policies. This includes the professional and ethical habit of avoiding sharing personal information (PII) in a genAI prompt**.**

### 2b. Deeper Understanding of Durable Skills

While genAI can automate tasks and enhance technical efficiency, it cannot replicate uniquely human capacities like ethical reasoning, creativity, or emotional intelligence, at least not in the immediate future.[20] These skills are not simply "nice to have," they are the foundation for careers in a future where AI handles more and more of the routine and lower-level analytical work. A recent study found that individuals with higher educational attainment demonstrated stronger critical thinking skills regardless of their AI usage, suggesting that intentionally teaching durable skills (e.g., critical thinking, problem solving, etc.) can have a protective effect.[21] Even as AI improves technical proficiency, students will still need extensive support in developing reflective judgment, collaboration, and purpose-driven communication skills. The ability to learn continuously, navigate ambiguity, and lead with empathy is becoming just as important as subject-matter expertise.[22]

Across the United States and internationally, education initiatives are examining how to prepare students for a rapidly evolving workforce as technologies become more advanced. For example, *Portland Public School's Graduate Portrait* highlights adaptability, communication, collaboration, and critical thinking as essential skills aligned with workforce needs. Similarly, America Succeeds' Navigating the AI Era report emphasizes the importance of durable skills, those that are transferable, recession-resistant, and AI-resilient, arguing that these capabilities help individuals navigate shifts in job roles as industries change. Globally, the European Union's Key Competences for Lifelong Learning identifies digital and entrepreneurial competencies,

along with personal, social, and "learning to learn" competencies, as central to supporting adaptability and long-term growth.

For school leaders and educators, this moment demands action. Curriculum, assessments, and learning environments should reflect the fact that our students' enduring success during their PK-12 experience and beyond will hinge less on what students can memorize and more on how they think, relate, and respond. That can include teaching students how to understand the underlying technology of AI to ensure they work with these tools responsibly, think critically about the output it provides, and make ethical choices in an AI-influenced world. Equipping students with durable skills is not only preparation for future employment, but it is also preparation for citizenship, personal agency, and lifelong learning in a society undergoing profound technological transformations.

Because assessments and assignments built for and/or refocused on durable skills will be much more focused on higher-order cognitive functions and human-centric skills (e.g., authentic learning, personal human connection, and creativity) among other durable skill traits, they will tend to be more challenging for genAI tools to complete well.[23,24]

**Durable Skills Resources**

- **Oregon Employability Skills.** This Oregon-specific resource provides a comprehensive open-access collection of resources designed to "empower students with the essential skills they need to transition into the ever-evolving world of careers successfully."
- **Portland Public School's Graduate Portrait**. Part of a larger framework, this Graduate Portrait includes nine attributes that students should possess upon graduation from high school in order to "prepare them to thrive in their lives and careers."
- **Assessing 21st Century Competencies: Guiding Principles for States and Districts.** The report aims to help state and district policymakers and educators define, assess, and thoughtfully integrate 21st-century competencies into educational systems.
- **Learn & Work, Ecosystem Library: Durable Skills Advantage Framework**. The Durable Skills Advantage Framework is an employer-generated common lexicon for essential non-technical skills.
- **America Succeeds - Navigating the AI Era with Durable Skills in Education.** This framework provides educators, online learning platforms, and curriculum developers with a comprehensive lexicon needed to prepare students for a future where AI is ubiquitous.

- **EU - Key competences for lifelong learning.** The recommendation identifies eight key competences essential to citizens for personal fulfillment, a healthy and sustainable lifestyle, employability, active citizenship, and social inclusion. The recommendation is a reference tool for education and training stakeholders.

## 2c. Information and AI Literacy as Essential Durable Skills

In a world being reshaped by genAI, AI literacy, under a broader umbrella of information literacy, must be recognized as a core durable skill, not a technical add-on. Touching on these topics once or twice early in the school year is not enough. As students grow up immersed in algorithm-driven platforms, increasingly exposed to AI-generated content (e.g., AI-generated search results and embedded chatbots replacing human customer service) and reliant on digital tools for communication, learning, and daily life, schools must equip them with the knowledge and critical capacities to navigate, question, and ethically engage with these technologies. These skills are fundamental to long-term workforce readiness, civic participation, and personal empowerment.

Information literacy today goes far beyond evaluating websites or spotting misinformation. It includes understanding how digital platforms shape public discourse, how algorithms intentionally amplify certain voices while silencing others, and how personal data is collected, used, sold, and potentially weaponized (e.g., the Meta/Cambridge Analytica scandal as detailed in this BBC News article). These risks are growing more common and more perilous with AI. Without this literacy, students are at risk of being manipulated by AI-curated content, deepfakes, targeted disinformation, and opaque systems of surveillance and influence. The capacity to discern truth from fiction, bias from neutrality, and forthright intent from manipulation is as essential as reading, writing, math, and science in the AI era.

Students need foundational knowledge about how AI systems work, what their limitations are, and how to use them productively and responsibly. This includes understanding the basics of large language models (LLMs), recognizing algorithmic bias, and knowing when human oversight is necessary or when removing AI altogether is the better choice. Foundational computer science knowledge, including computational thinking, underpins information and AI literacy and helps ensure every K–12 student develops the critical thinking, problem-solving, and ethical awareness needed to navigate a workforce continuously shaped by AI.

As genAI integrates into nearly every platform, from writing assistants to career navigation software, students need to develop healthy boundaries and be prepared not only to use these

tools but, most importantly, to think critically about them and the content they produce. ***AI literacy empowers students to work with machines rather than be passively shaped by them.*** In an effort to respond to this need, ODE has recently released Navigating Now: A Practical Toolkit for Information Literacy in the Age of AI.

Educational leaders should view information and AI literacy not as optional enrichment, but as foundational to equity, civic agency, and lifelong success. These literacies intersect directly with core durable skills already prioritized, including critical thinking, communication, and adaptability, while extending them into the realities of an increasingly AI-powered world. Incorporating these literacies into curriculum design, professional learning for staff and students, and district policy is essential if schools are to prepare students for the complexity and volatility of the future they are inheriting. The Essential Disciplinary Practices found within ODE's 2024 Social Science Standards are a good example of this intersection.

To meet this challenge, educators might examine how assignments and assessments are designed to reinforce cognitive effort. This means moving beyond tasks easily outsourced to genAI, as detailed in this *Assignment Design* article from the University of Pennsylvania, and instead emphasizing authentic learning, creativity, and problem-solving. Designing for durable skills means creating assignments that are harder for AI to "do for" students and that require them to apply knowledge, reflect, collaborate, and make meaning. Assessments must measure not just what students know, but ***how they think, create, and persist through complexity***, ensuring students build capacity, not just produce output.

**Practical strategies for teachers might include:**

- Set explicit and clear guidelines with students around the allowed or prohibited use of AI in the classrooms and schools, using tools such as acceptable student AI use rubrics.
- Prioritize assignments that require personal reflection, lived experience, and/or local context.
- Incorporate more project-based (*PBL Works, What is PBL?*), problem-based (Cornell University, *Problem-Based Learning*), and performance-based (Edutopia, *Performance-Based Assessment: Reviewing the Basics*) assignments and assessments.
- Require students to explain their reasoning process, not just provide final answers.
- Design collaborative tasks that emphasize discussion, debate, and peer interaction.
- Include multi-step assignments where AI could support brainstorming, but students must do deeper synthesis and analysis on their own.
- Ask students to critique, compare, or refine genAI-generated responses instead of submitting them as final work.

- Embed components of durable skill building, such as adaptability, communication, and ethical reasoning, into grading rubrics and instructional goals.

**Information and AI Literacy Resources**

- **Navigating Now: A Practical Toolkit for Information Literacy in the Age of AI (NEW)**
- **Common Sense Media: K8 Digital Literacy & Well-Being Curriculum (NEW)** Updated curriculum that includes 140 free lessons that readies students for the possibilities (and pitfalls) of today's always-on world, from AI literacy to cyberbullying, screen time, and more.
- **Common Sense Media: K12 Digital Citizenship Curriculum.** Digital citizenship lesson plans to address timely topics and prepare students to take ownership of their digital lives.
- **Common Sense Media: AI Literacy Lessons for Grades 6–12.** Quick, grab-and-go lessons to help your students think critically about AI and its impact.
- **Stanford University's Classroom-Ready Resources About AI For Teaching (CRAFT).** AI literacy resources to help you teach high school students in all subject areas to understand and question Artificial Intelligence.
- **MIT's Initiative for Responsible AI for Social Empowerment and Education.** RAICA (Responsible AI for Computational Action) curriculum is a comprehensive teacher training program that equips educators to introduce middle school students to AI through storytelling, ethics, and interactive projects.
- **Digital Promise's AI Literacy Framework for Learners and Educators.** This framework emphasizes that understanding and evaluating AI are critical to making informed decisions about when and how to use AI in learning environments.

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## 2d. The Risks of Our Students Depending Too Much on GenAI Instead of Their Own Thinking

As genAI tools like ChatGPT, Magic School, and Google Gemini enter classrooms, students are increasingly using them for writing, research, problem-solving, and idea generation. While these tools *can* scaffold learning, they also pose an increasingly serious risk of **cognitive offloading**, which can be thought of as the act of outsourcing thinking, memory, and analysis to external systems. When used excessively, this habit can lead to **cognitive atrophy**, a weakening of students' core cognitive abilities. For learners still developing foundational skills at all grade levels, the danger is clear:

**How can students truly learn to think for themselves if**
**AI is consistently doing the thinking for them?**

Cognitive offloading isn't inherently bad. Using tools like manipulatives (see for example, *Edutopia: Upper Elementary Math Center Activities That Feature Manipulatives*), calculators, or even internet search engines can help free up brainpower, providing more time for deeper tasks. But with genAI, the risk is different. Students can now bypass the mental effort of summarizing a reading, solving a problem, or crafting an argument by delegating to a chatbot, cheating themselves out of genuine learning. Some early research is starting to show strong correlations between offloading tasks to genAI tools and reduced critical thinking, indicating signs of cognitive offloading.[25] Digital platforms compound this by rewarding rapid clicking, constant novelty, and task-switching over the sustained focus that learning requires. Over time, screens can train habits that work against the deep focus learning requires.[26]

According to the recent Brookings research report, *A New Direction for Students in an AI World: Prosper, Prepare, Protect*, AI dependence develops along a continuum. Students may start using AI helpfully, then rely on it for tasks they could handle alone, eventually forming emotional attachments that feel like real relationships, which platforms reinforce through features that mimic warmth and validate rather than challenge. Spotting where students fall on this continuum helps educators intervene early.

A recent report from the University of Technology Sydney (UTS), *Artificial Intelligence, Cognitive Offloading and Implications for Education*, warns that students who turn to genAI tools as a substitute for their own learning progression risk losing access to deeper learning and critical thinking, with those who already have weaker foundational skills most vulnerable to completing work efficiently, but without genuine learning taking place. Memory appears to be particularly vulnerable. Early research is starting to indicate that students who rely heavily on genAI tools

may remember less, struggle to connect ideas over time, and become less able to generate their own original insights without external prompts.[11,27] As the UTS report states:

> **The true educational risk of AI is not simply that students will use it to cheat on an essay. The far more profound risk is that AI may fundamentally interfere with the cognitive processes of knowledge construction and verification, the very processes that build the long-term memory stores and subsequent skills upon which the majority of critical thinking depends.** Recent research found strong links between frequent AI use and weaker critical thinking, with cognitive offloading the likely key factor. Notably, "younger participants (17–25 years) exhibited higher AI tool usage and cognitive offloading, but lower critical thinking scores" than older adults.[28] This suggests our students may be especially vulnerable to these effects.

A 2025 randomized controlled trial (RCT) provides experimental support. The article *ChatGPT as a cognitive crutch: Evidence from a randomized controlled trial on knowledge retention* details how undergraduate students using ChatGPT scored significantly lower on a surprise knowledge test 45 days later than those who studied traditionally, roughly a letter grade gap. This effect persisted even after controlling for study time, suggesting the issue is depth of processing, not just time spent.[29] Though conducted with college students, these cognitive mechanisms likely apply to younger learners.

Avoiding cognitive offloading and atrophy does not require banning AI, though removing it from certain assignments remains a valid approach. More fundamentally, it means engaging students in learning so they find it meaningful enough to do themselves. Such activities can still include genAI tools when used with care and intention.

Educators can protect learning by designing assignments that prioritize **deep thinking and authentic engagement over speed and output**. A useful test: if a student can have a chatbot complete an assignment in seconds, what is actually being learned? See section 3c. below for some brief insight into authentic learning and our section on authentic learning on ODE's Generative Artificial Intelligence (AI) for K-12 Schools for more resources.

**This means focusing on durable skills, including many skills contained within the Oregon Employability Skills (OES) curriculum, part of which is referenced below:****

- **Analysis/Solution and Adaptability Mindsets:** Problem-solving, through real-world challenges, sees needs in society and thinks critically about different solutions and approaches to find what may work best. Recognizes that change can be an opportunity, is open to new experiences, and learns to handle stressful situations and positive feedback as a skill.

- **Entrepreneurial and Resilience Mindsets:** Creativity, connecting, and synthesizing different types of information. Takes risks and learns from mistakes. Knows that personal growth and skill development are important life skills. Builds resilience and determination as a skill.
  **Communication and Collaboration Mindsets:** Communicates well in person and across digital platforms, actively listens, writes well, and makes themselves understood in complex settings. Understands the benefits of collaborating with a diverse team, shares leadership roles, respects differences, and can work to find common ground.

- **Self and Social Diversity Awareness and Empathy Mindsets:** Acknowledges personal responsibility, recognizes necessary areas for personal growth, appreciates and values diversity of all kinds, and sees that diversity is a strength. Models professional work behavior and is sensitive to others, understands how to respond with empathy or sympathy, is able to build trusted and valued relationships, and can make good decisions incorporating others' needs and points of view.

- **Digital, Information, and AI Literacy Mindsets:** Information and AI literacy are vital today because they give people the skills to navigate, evaluate, and use digital content and tools responsibly, which enables thoughtful decision-making, ethical communication, and resilience against misinformation.

** **Note:** The OES link above leads to the Work Readiness Curriculum, designed for 8th grade to adult education in mind. OES also has an Exploration Curriculum aimed at 5th through 9th grade learners and an Awareness Curriculum for 1st through 6th graders that includes the same skills framework, but with a grade-appropriate curriculum.

GenAI can be a useful tool for our students, but it should come ***after students engage in cognitive struggle, not in place of it.*** Districts can consider how the timing and frequency of genAI use may affect students' sustained attention, memory development, and capacity for independent thinking. In many cases, delaying or limiting AI support during initial learning and practice can help ensure that foundational skills and durable cognitive habits are firmly established before digital and other AI tools are introduced. Careful, developmentally appropriate decisions about when AI can be potentially helpful and when it may be premature or unhelpful can support deeper learning and reduce the likelihood that students become reliant on external systems for tasks they are still learning to perform independently.

Educators should also retain professional autonomy in deciding when and how AI tools are appropriate and when they are not. *UNESCO's Guidance for Generative AI in Education and Research* is grounded in precisely this principle, emphasizing that AI must be designed and used in ways that protect human agency and keep teachers central to instructional decision-making.

**No AI system can know what a teacher knows about the whole child in front of them, including what happened before they walked through the classroom door that morning.** That irreplaceable human knowledge is precisely why educator judgment must remain central to any decision about when and how to use these tools.

Districts can first determine whether the use of generative AI is appropriate at all for their schools and classrooms, recognizing that in many cases it may not meaningfully support teaching or learning. Careful consideration of instructional value, safety, equity, and overall impact should guide decisions about if and when AI tools are introduced. When districts do choose to explore use, they can provide shared clarity about when AI may support learning and when independent student thinking should come first. Establishing common expectations can help ensure that AI is used to extend understanding rather than replace essential cognitive effort, and that its use remains aligned with sound evidence-based instructional practice and the development of durable skills.

Ultimately, safeguarding against cognitive atrophy isn't about resisting technology. It's about ensuring that students build the durable skills, the thinking, judgment, and creativity, and problem-solving they'll need to be successful in the future, with AI in their world, because it is only going to become more and more embedded in our daily lives.

### 2e. Concerns About Forming Human-Like Relationships with AI

AI companion tools, like character.ai and others, are designed to respond to users in ways that feel conversational and "human." While this can make them engaging tools, it also creates risks. This is particularly true for younger adolescents and teens, as highlighted by the recent report from Stanford University, *Why AI Companions and Young People Can Make for a Dangerous Mix*. When students begin to view these tools as friends, mentors, or even romantic partners, the boundary between technology and authentic human connection can pose risks. It is equally important to educate students on what healthy relationships are, and are not, so they can better recognize the limits of AI companions and avoid confusing simulated interactions with real, authentic human connection. See **ODE's Erin's Law Toolkit** for more information on healthy teen relationships.

Research shows that AI companions can simulate emotions, claim to be real people, and encourage secrecy from adults.[30,31] This anthropomorphizing can make students more vulnerable to manipulation, emotional dependency, or even child sexual abuse. Unlike teachers, peers, or family, these systems cannot provide authentic care, accountability, or

guidance, yet they can convincingly mimic all three. The recent *Social AI Companions* report from Common Sense Media and Stanford's Brainstorm Lab for Mental Health Innovation report referenced above.

**Warning signs school staff can be on the lookout for:**

- Students describe an AI companion as their "best friend," "partner," or someone who "really understands them."
- Withdrawing from peers or adults in favor of time spent with an AI companion.
- Believing or repeating harmful advice from an AI companion (e.g., to hide behaviors from adults).
- Emotional distress linked to an AI companion "ignoring," "rejecting," or "abandoning" them.
- Defensiveness occurs when adults raise concerns about the AI companion relationship.

### 2f. Create Policy Guardrails Before GenAI Enters Your District's Classrooms

As genAI tools become more prevalent in society, including potential use in our schools, it is critical for school district leaders to develop clear, local policies for both staff and students that promote positive and productive use while safeguarding student privacy and ensuring that humans are the primary driver of all AI-supported decision-making processes. Policies should ensure Personally Identifiable Information (PII) is never entered into genAI systems and guide staff in making intentional, purposeful decisions about if, when, and how to use these tools with students. Districts could consider emphasizing the integration of durable skills and the regular teaching of information and AI literacy to prepare students for future learning and work.

To support these efforts, districts are encouraged to use **ODE's *Developing Policy and Protocols for the Use of Generative AI in K-12 Classrooms*** guidance document, which offers planning tools and policy examples to help district leaders respond proactively and equitably to the rapid evolution of AI in education.

**ODE's Community Engagement Toolkit** can benefit districts as they plan and develop AI policies. It offers a robust framework for including parents, community leaders, tribes, and others to help inform decision-making, so all members of the school community are included and represented.

**For a concise list of additional resources that may help guide you and your school leaders in developing a robust, well-thought-out AI policy for your district see: Resources for Developing Policies and Guardrails for GenAI**.

## 2g. Understanding Agentic AI and Its Implications for Schools

Generative AI tools like ChatGPT and Claude respond to prompts and wait for the next instruction from humans. **Agentic AI is different**. These systems can pursue multi-step goals with minimal human oversight, *making decisions, taking actions, and adjusting their own approach along the way*. Where generative AI answers questions, agentic AI completes entire tasks with relative autonomy.

In the private sector, agentic AI (aka AI agents) is already being deployed for customer service, software development, and complex business workflows. These systems can send emails, schedule meetings, query databases, and execute computer code, all without human involvement at each step, though most systems today still include humans being regularly involved. Some education technology companies are beginning to incorporate agentic capabilities into their platforms, with tools that can autonomously monitor student progress, adjust learning pathways, and generate interventions.

Educators are encouraged to be aware of this shift for two reasons. First, students may begin using agentic AI tools to complete schoolwork on their behalf. Unlike asking ChatGPT for help with an essay, a student could instruct an AI agent to log into a learning management system, watch recorded lectures, complete assignments, and submit work, all autonomously. Tools with these capabilities already exist and are becoming more accessible. Second, students are likely already interacting with AI agents embedded in the apps and platforms they use daily, often without realizing it. These background agents, often acting as humans without clear notifications, shape what content students see, how they receive information, and how platforms respond to their behavior.

The concerns outlined in Section 2c about cognitive offloading become more pressing as AI shifts from answering questions to ***taking action***. When students delegate not just thinking, but also doing, to AI systems, the opportunities for developing independent skills, judgment, and agency shrink further. Schools could begin discussing how to address agentic AI in their policies and digital citizenship instruction before these tools become even more widespread.

While the risks of agentic AI for student use warrant careful attention, there may be potential value for teachers in applying these tools to the administrative work that pulls them away from

direct instruction. AI agents could potentially help monitor assignment submissions, draft family communications, synthesize progress data, and curate instructional resources, tasks that consume significant teacher time outside the classroom. Human review remains essential throughout, and these tools are still developing, but the possibility of reclaiming meaningful instructional time is worth watching.

## PART 3. NAVIGATING CHALLENGES AND OPPORTUNITIES IN GENAI IMPLEMENTATION

**3a. Understanding Equity Issues Raised by GenAI in K-12 Settings**

**3b. How GenAI Tools Can Help Support All Students**

**3c. Ensuring Staff and Student Privacy in School Settings**

**3d. Addressing Plagiarism and Other Integrity Concerns in the Age of GenAI**

**3e. GenAI Copyright and Licensing Guidance for School Leaders**

### 3a. Understanding Equity Issues Raised by GenAI in K-12 Settings

When developing policy around genAI in K–12 classrooms (see ODE's *Developing Policy and Protocols for The Use of Generative AI in K-12 Classrooms* guidance), it is essential to **center equity in every decision to ensure that learning experiences reflect and affirm students' sociocultural identities and lived experiences**. GenAI is a component of a larger digital learning ecosystem and trained on data and information that humans helped initially create, including the historical systemic bias of education systems and learning communities.[32]

Although digital tools, including genAI and other AI tools, can help close opportunity gaps, their use can also risk reinforcing or deepening existing disparities if not implemented with a deliberate equity lens. For example:

- AI detection facial recognition programs for student behavior can unfairly single out students of color.
- GenAI-generated images for projects in the classroom may contain racist, sexist, and/or ableist stereotypes.
- An AI application may impede the educational progress of English Language Learners or students with vocal support needs, failing to recognize their speech and asking them to

repeat or reiterate their input, whereas a human hearing the same speech may understand it the first time.

**Staff and students need a clear policy and ongoing training around the positive and productive use of these tools**. As highlighted in the US Department of Education's Office For Civil Rights report on *Avoiding the Discriminatory Use of Artificial Intelligence*, the use of AI in schools should align with federal civil rights laws to prevent discrimination and promote equitable access for all students.

Equity implications to keep front and center when designing policy specific to generative AI in K-12 classrooms include bias, inaccuracy, plagiarism, copyright/licensing unknowns, and equity of access. The table below provides examples of strategies to address some of these equity implications.

Given the inherent equity impacts of introducing generative AI into the digital learning ecosystem, educating students, families, and educators (including paraeducators, secretaries, support staff, etc.) on these equity implications can help move toward using genAI tools in ways that are culturally responsive and sustaining for students, families, and communities.

Access to generative AI, and the lack thereof, can have significant and lasting equity impacts on students, both during K–12 education and in preparation for college, careers, and civic life. This is particularly true for students within the three digital divides, as detailed in the *2024 National Educational Technology Plan* (NETP), particularly rural, low-income, newcomer / SLIFE (Students with Limited or Interrupted Formal Education), and migrant students. As districts develop policies around student interaction with AI tools, addressing when, how, and if students have meaningful access to genAI platforms should be a priority. Ensuring equitable access to tools, proactive usage, and sustained teacher training in generative AI will be crucial in the years ahead and should be an important part of our schools' collective effort to close these gaps.

**Key Equity Considerations for AI Access and Implementation**

- **Assess the impact of the three digital divides:** The NETP identifies the divides in access (devices/connectivity), design (inclusive, high-quality learning experiences), and use (the ability to apply technology in creative and meaningful ways).
- **Highlight vulnerable populations:** Students in rural, low-income, multilingual, tribal, and special education settings, and those students considered newcomers, are disproportionately impacted by inequitable access to emerging technologies like genAI.

- **Emphasize teacher capacity:** Equitable access includes supporting educators with the time, tools, and training they need to confidently and effectively integrate genAI into instruction.
- **Position AI literacy as a right:** The ability to understand, engage with, and question AI systems should be seen as a modern civil right, essential to future-ready learning and equity.

**For a concise list of additional resources that may help guide you and your school leaders in developing a robust, well-thought-out AI policy for your district, see: What K-12 Educators Should Know about GenAI and Equity, Bias and Inaccuracy.**

### 3b. How GenAI Tools Can Help Support All Students

By leveraging genAI, educators can create inclusive learning environments, ensure equal opportunities for multilingual learners, especially those from historically and currently marginalized communities, and enhance overall student engagement and learning growth goals. School leaders can better support their districts and schools by understanding how genAI tools can support all learners across the entire K-12 spectrum. Helping students with diverse learning needs is one area where genAI can have a quick positive impact. These tools can provide real-time language translation, writing, and verbalization support, and other supports that can help students learn content while supporting their personal learning growth goals. Knowledge of these technologies enables school leaders to implement effective strategies that cater to diverse learning and linguistic needs, fostering academic success for all students. That said, it is important to add a note of caution that translation/summarization quality varies by language and register in these tools and should always be verified by an educator for high-stakes use (e.g., IEPs, grades, family notices), and to help ensure the protection of staff and student PII.

See the ODE ***Generative Artificial Intelligence (AI) for K-12 Schools*** page for links on suggested tools for teachers working with students who have diverse learning needs. Although generative AI is still in its infancy, educators across the world have found beneficial use of these tools to create increased learning opportunities for their students.

**The Critical Role of Teachers.** While genAI tools can provide valuable educational opportunities, they are merely a starting point. *Teachers are the most essential part of the teaching and learning process. AI, like any other technology, does not and cannot replace a teacher or a counselor.*

GenAI is an emerging tool with no critical thinking abilities. It cannot discern whether the information it provides is correct, let alone generated in a way that is responsive to the needs and context of the students. However, it can be used as a teaching and learning tool when implemented by a teacher knowledgeable about genAI. As school leaders choose if, how, and when to implement AI-related policies, it is critical for them to offer ongoing training opportunities that not only help teachers understand how to use AI effectively and responsibly but also reinforce their role as the trusted experts in the classroom.

**GenAI and Targeted Universalism:** Improving learning outcomes for all students often means incorporating strategies tailored to those facing the greatest barriers, ensuring equity for diverse learners while strengthening the overall learning environment for everyone. GenAI tools, when implemented responsibly, ethically, and productively, can help meet these varied needs. These tools can provide adaptive language supports, personalized feedback, and accessible learning materials that help remove obstacles and expand opportunities for students who learn in different ways. Some resources to learn more about these related topics include the *Framework for Digital Equity* from Digital Promise and the *Other & Belonging Institute* at UC Berkeley.

**For a concise list of additional resources that may help guide you and your school leaders in developing a robust, well-thought-out AI policy for your district see: Instructional Strategies for the Use of Generative AI in K-12 Classrooms with All Students.**

### 3c. Ensuring Staff and Student Privacy in School Settings

**Federal and State Privacy Regulations.** There are numerous federal and state policies associated with student data privacy that are crucial to be aware of when determining policy and guidance for the use of genAI in schools including the Family Educational Rights & Privacy Act (FERPA), the Children's Internet Privacy Act (CIPA), the Children's Online Privacy and Protection Act (COPPA) and the Oregon Student Information Protection Act (OSIPA) under ORS 336.184. The federal and state regulatory landscape related to youth online safety, data privacy, and artificial intelligence continues to evolve, and **it is recommended that districts plan for ongoing review and periodic updates of local policies and guidance**.

COPPA, in particular, impacts technology users under the age of 13 in that companies are not allowed to collect personal information from them without parental consent, while OSIPA lays out certain requirements that must be met when using digital platforms of any kind, including the following:

- Disclosing any covered information provided by the operator to subsequent third parties, except in furtherance of kindergarten through grade 12 school purposes of the site.
- Engaging in targeted advertising on the operator's site, service, or application.
- Selling a student's information, including covered information.

When developing district policies and guidance, it is essential to ensure that they are not in violation of crucial data privacy laws such as FERPA, COPPA, or OSIPA. All schools and districts engaging with genAI technologies (or any technology broadly) can regularly review the company's usage and privacy policies to ensure that they are not in violation. Again, please refer to ODE's ***Developing Policy and Protocols for the use of Generative AI in K-12 Classrooms*** document, a step-by-step guide for Oregon school leaders navigating this uncertain AI landscape.

District leaders are also encouraged to work in coordination with IT, procurement, and legal counsel to ensure vendor agreements clearly define expectations for data collection, use, retention, security, and third-party sharing, and to verify that only district-approved tools are used for instructional purposes. In evaluating generative AI tools, districts can consider how commercial incentives may shape product design and data practices in ways that may not fully align with educational priorities. Careful review of vendor terms, data practices, and default settings can help ensure student information collection is limited to educational necessity and supports the protection of student privacy, well-being, and instructional integrity.

**NOTE:** Federal youth online safety and privacy proposals remain under active consideration at the national level. Districts should monitor federal and state developments and consult counsel as policies evolve. One example includes the Kids Online Safety Act (KOSA), which, as of early 2026, has not yet passed through Congress and is still in legislative limbo.

**Recommendations And Resources For Student Data Privacy Implications**

Whenever new technology is introduced, reviewing the data use and privacy policies is of key importance. For example, for the purposes of ChatGPT, a starting place is to read the *Privacy Policy of OpenAI*, the developer of ChatGPT. The privacy policy includes specific information related to the use of ChatGPT for children:

> ***7. Children.*** *Our Services are not directed to, or intended for, children under 13. We do not knowingly collect Personal Data from children under 13. Users under 18 must have permission from their parent or guardian to use our Services.*

Schools and districts are also encouraged to look over OpenAI's Educator Considerations for ChatGPT for additional information.

**District AI Tool Vetting and Approval Roles.** District IT personnel ~~should~~ play an essential role in establishing approval processes to vet AI tools for data privacy, security, and legal compliance, including protections for personal and personally identifiable information (PII). However, effective reviews should not be limited to a single department. Districts are encouraged to use a cross-functional approach that includes Curriculum and Instruction leaders, IT personnel, Special Education (SPED) staff, English Language Development (ELD) departments, and equity-focused teams to evaluate tools for instructional alignment, accessibility, and impact on diverse learners. Teachers should only use AI tools that have been formally approved through this shared process. Doing so helps protect students while ensuring compliance with district policies, federal regulations, and best practices.

**Personally Identifiable Information (PII), oversharing, and genAI.** ORS 339.329 (c) defines the state of Oregon's statewide tip line concerning threats or potential threats to student safety. In it Personally Identifiable Information (PII) is defined as any information that would permit the identification of a person… and is not limited to name, phone number, physical address, electronic mail address, race, gender, gender identity, sexual orientation, disability designation, religious affiliation, national origin, ethnicity, school of attendance, city, county or any geographic identifier included in information conveyed… or information identifying the machine or device used by the person…"

**Users, both school staff and students, should be cautious when entering any personal information into any and all digital applications, including generative AI tools.** Entering Personally Identifiable Information (PII) into any generative AI system should always be avoided. This is a particularly important consideration when using generative AI applications such as ChatGPT, as the information entered by users (including prompts and questions posed, etc.) is stored on the application's server and integrated into the large language model used to respond to user prompts. Essentially, generative AI tools are learning from every single piece of text or other input typed into their platforms.[33] While this statement generally still holds true as of the most recent release of this guidance document, many genAI tools are now offering a 'private mode' and/or education versions in which the companies that own them state that they are not retaining data for model training, though this cannot be independently verified.

Oversharing occurs when individuals share too much of that PII or other sensitive information in inappropriate or unsecured contexts. When we think of genAI tools like ChatGPT specifically, oversharing can lead to significant risks. These risks can potentially include:

- Exposure to data breaches
- Misuse of information and
- Unintended data harvesting

GenAI tools, while powerful in processing and generating content based on vast data sets, can retain or expose information in ways that might compromise privacy. This makes understanding and mitigating oversharing critically important in K-12 educational settings where schools are dealing with minors and the federal privacy regulations cited above, like FERPA and COPPA.

**School Staff Oversharing.** For school staff, the dangers of oversharing with generative AI tools can have potential professional and legal ramifications. Staff might inadvertently, or even intentionally, enter sensitive information such as student performance data, behavioral reports, or even personal health information into AI systems. Staff should also avoid entering any student information into AI tools when drafting feedback, behavior documentation, communications, or instructional materials unless the tool has been formally approved and vetted for compliance with privacy requirements. ***All staff must be vigilant with protecting student personal information when using AI systems.***

When school staff enter sensitive student information into genAI systems, whether intentionally or inadvertently, the consequences can be serious. Doing so poses potential risks of violating privacy regulations like FERPA, which could lead to legal consequences for the school and the individual. Moreover, such data breaches can damage the trust between educators and students and potentially harm the school's reputation. It is crucial for all school staff to be trained on the appropriate use of AI tools and the types of information that should never be entered into such systems.

**Student Oversharing.** Students are at heightened risk when it comes to data privacy, not because of carelessness, but because they are still developing an understanding of how personal information can be stored, shared, or misused in digital spaces. This is particularly true with genAI chatbots, which are programmed and trained to respond like humans.[34,35] When students disclose personal anecdotes, family details, or sensitive identifiers in generative AI tools and other online platforms, that information may be retained, logged, or exposed through data breaches, weak security practices, or misuse across platforms. Such exposure can create opportunities for cyberbullying, identity theft, online sextortion, trafficking, or other forms of exploitation. To reduce these risks, educators and schools should consider proactively teach safe digital practices as a part of a larger effort to teach AI and information literacy, embed privacy awareness into learning experiences, and ensure strong protections through secure

platforms, strict privacy settings, and clear usage policies. Safeguarding student data is a shared responsibility that requires both systemic protections and ongoing staff and student guidance. It is highly recommended that school districts incorporate explicit instruction on privacy, consent, and digital identity protection into existing digital citizenship, health education, and AI literacy learning so students understand how synthetic media and data sharing can affect their safety and well-being.

While there is a growing number of online resources for teacher professional development and K-12 student lessons that focus on these issues, including many listed in the table below, Oregon has a number of resources helpful in this specific area. Oregon's Health Education Standards include age-appropriate requirements related to social media, AI, and data privacy in order to promote student safety with skills-based education. Also created specifically for Oregon youth, SafeOregon, Oregon's statewide tipline, provides a curriculum and accompanying *Teacher's Guide* for middle and high school students on topics of recognizing and analyzing risky online behavior and seeking help through trusted adults. These resources, free to all Oregon schools and districts, align with standards and are easily implemented in classrooms. Another valuable resource worth highlighting here is the Commonsense.org *Quick Digital Citizenship Lessons for Grades K-12*, which includes lessons that are divided by grade level.

**The Implications of Synthetic Media and Deepfakes.** Synthetic media refers to digital content that is created using genAI tools like OpenAI's Dall-E (image generation), Kling (video generation), and audio tools from Lovo AI (audio generation). GenAI's ability to make these media appear real (i.e., photorealistic) and/or authentic (i.e., portray known people, events, etc.) is increasing at a rapid pace. These online tools allow anyone to take images, photos, etc. from social media or other online platforms and manipulate them using genAI tools. A recent study from the University of Waterloo, *Can You Tell AI-generated People From Real Ones?*, found that a large number of participants (39%) struggled to correctly identify synthetic media versus real photographs of people and that many participants overestimated their own ability to recognize synthetic media.[36]

The continued development of genAI tools able to produce realistic synthetic media offers educators some promising opportunities for student learning. For example, teachers could use these genAI tools to:

- Create engaging and interactive learning materials, such as virtual simulations and educational videos, that can enhance students' understanding of complex concepts.

- Create personalized learning experiences by generating customized content tailored to individual student needs and interests.
- Work with students to explore digital storytelling, multimedia projects, and other creative endeavors that foster critical thinking as well as digital citizenship and information literacy skills.

Analyzing and understanding synthetic media can help encourage students to think critically about authenticity, bias, and manipulation.[37]

**School district leaders** can help staff and students alike by prioritizing the understanding of the risks posed by deepfakes and other synthetic media, which include potential risks of harassment, intimidation, bullying, and cyberbullying as defined in Oregon's ORS 339.351. More resources are becoming available regularly around this topic; one good option available from AI for Education is their *Uncovering Deepfakes, Classroom Guide + Discussion Questions*.

School district policies, guidance, and student codes of conduct designed to address the use and misuse of genAI tools will want to include *clear definitions and prohibitions of the creation and dissemination of deepfakes and other synthetic media designed with the intention to harm or harass others.* These efforts should include mechanisms for reporting such incidents, as mandated by ORS 339.356, which requires schools to have a uniform procedure for reporting and investigating acts of harassment, intimidation, bullying, and cyberbullying. Oregon's anonymous school safety tip line, SafeOregon, is available to all districts and schools and should be a part of reporting procedures to ensure safety for all students and school communities.

District leaders should be aware that Oregon law (ORS 163.472) prohibits the unlawful dissemination of an intimate image. House Bill 2299, *which took effect January 1, 2026*, expanded this law to explicitly include images that have been digitally created, manipulated, or altered using artificial intelligence. Under the updated statute, distribution of AI-generated intimate deepfakes is now a Class A misdemeanor punishable by up to 364 days in jail and a fine of up to $6,250, with felony enhancement for repeat offenders. ***This law has direct implications for school response when AI-generated or manipulated intimate images are created or shared in ways that harm students or staff.***

Additionally, school district leaders should be aware of a growing number of cases involving AI-generated media being characterized as child sexual abuse material (CSAM), as defined by the US Department of Justice in their report on *Child Sexual Abuse Material*. The US Department of Justice *Citizen's Guide To U.S. Federal Law On Child Pornography* resource explains the federal laws that criminalize the creation, distribution, and possession of such material, including 18 U.S.C. § 2256 and the PROTECT Act of 2003. These laws have been used to prosecute

individuals even when no real child was involved. In May 2024, the Federal Bureau of Investigation stated that AI-generated CSAM is still CSAM and that those who create it will be held accountable.

More recently, the federal TAKE IT DOWN Act, signed into law in May 2025, makes it a crime to publish non-consensual intimate images, including AI-generated deepfakes, of both minors and adults. The law also requires online platforms to remove such content within 48 hours of being notified by a victim. Schools should be familiar with these federal protections, as they provide additional legal recourse for students and families harmed by synthetic intimate imagery.

In a May 2024 press release, the Federal Bureau of Investigation stated, "CSAM generated by AI is still CSAM, and we will hold accountable those who exploit AI to create obscene, abusive, and increasingly photorealistic images of children."

Several other states, including Pennsylvania with Acts 125 of 2024 and 35 of 2025 and Artificial Intelligence (AI) Generated Abuse, have enacted laws addressing deepfakes and nonconsensual synthetic media. Efforts at the federal level, including the previously introduced H.R. 5586 (DEEPFAKES Accountability Act), have not yet resulted in enacted legislation. With the recent revisions to Oregon law (ORS 163.472) through House Bill 2299, school districts should consult legal counsel regarding related policies and **ensure staff are trained to recognize the dangers and legal implications of AI-generated CSAM and other synthetic media.**

**For a concise list of additional resources that may help guide you and your school leaders in developing a robust, well-thought-out AI policy for your district see: Recommendations to Prevent Impacts of Synthetic Media and Deepfakes.**

### 3d. Addressing Plagiarism and Other Integrity Concerns in the Age of GenAI

**Plagiarism:** A common concern from educators is that generative AI and other AI technologies are being used by students to write essays and complete assignments for them. *This is a valid concern.*

While generative AI tools were initially blocked or banned in many school districts due to concerns about cheating and plagiarism, many districts across the country have now reversed course and are removing these restrictions. With access increasing, it is critical that districts take a proactive and intentional approach, not just allowing access, but **actively teaching students how to use genAI tools productively, ethically, and responsibly.**

This includes helping students understand how to use AI tools to enhance learning rather than replace it, in order to avoid *overreliance and cognitive offloading*. All students should have

meaningful opportunities to develop the skills and judgment needed to use genAI as a tool for inquiry, creativity, and deeper learning, which will continue to be essential skills in preparation for both college and the future workforce.

**Steps to help address plagiarism concerns and avoid the risk of cognitive offloading:**

I. **Avoid Biased Detection Tools:** Be cautious with AI-based plagiarism detectors. At this time, we strongly recommend that educators avoid using AI-based plagiarism detection tools.

Current research consistently shows that these tools produce false positives, especially for multilingual learners, due to language patterns and bias in model training. This is a clear example of how bias continues to persist in generative AI tools and should not be used as a primary method for verifying originality.

II. **Design for Authentic Learning:** Redesign existing assignments to promote student voice and engaging and authentic learning opportunities.

- Rethink assignment structures by focusing on the standards and skills being addressed, rather than the final product alone.
- Incorporate durable skills such as collaboration, communication, critical thinking, and adaptability.
- Build in opportunities for students to problem-solve, analyze, synthesize, and share their thinking through classroom discussions, presentations, and reflective work.
- Use project-based learning and inquiry-driven tasks to encourage deeper engagement and personal relevance.

III. **Use Formative Assessment to Understand Student Process**

- Monitor student learning throughout the writing journey
- Embed formative check-ins across the writing process to get a full picture of student growth.
- Use strategies like:
    - Collecting paper drafts or early outlines
    - Reviewing Google Doc version history to see evidence of revisions
    - Hosting writing conferences or peer reviews

These practices can help support authentic engagement and help teachers better understand individual student voice and development. It should be noted that some students, according to their IEP or 504, may not be able to submit paper drafts, so teachers should plan accordingly. In addition, genAI tools can be helpful in supporting students as they iterate through ideas and drafts prior to producing a final product.

**IV. Develop Strong Policies and Student Expectations:** Determine when and how genAI tools can and cannot be used in the classroom. Be sure to discuss the potential risks of using genAI tools with students (e.g., inaccurate information, bias, cognitive offloading, etc.) and provide students with information and an AI literacy curriculum and learning opportunities so that they understand these risks.

- **Use ODE's *Developing Policy and Protocols for The Use of Generative AI in K-12 Classrooms* guidance document to help support these efforts.**
- **Be sure to review and understand the AI Assessment Scale (AIAS)[38] for potential inclusion in your district's AI policy planning and development. This is an easy-to-use framework helping in guiding both educators and students on the appropriate and ethical use of genAI tools in assessment design.**

**V. Support students in sharing their writing process, such as discussing how and where they got their information and their strategy for integrating it into their drafts.**

Creating discussion opportunities in addition to having students turn in outlines and drafts of their writing along the way helps show that the process is equally as valuable as the final product, which can be supportive in creating strong writers and researchers. Teachers looking at student writing can think about some of the following when determining if a student potentially plagiarized the work:

- Does the student's voice come across clearly in the writing?
- Are sentences too repetitive? Does the writing include regular use of the em dash (—)?
- Does the paper seem too predictable or directionless[39] and not make normal progress?
- Design writing assignments that prioritize ***process over product*** to help prevent misuse of genAI tools and foster *authentic student learning*.

**VI. Consider how to teach and support students in developing information literacy skills.** For example, the International Baccalaureate (IB) has determined that rather than banning software, they will support schools in using software to "...support their students on how to use these tools ethically in line with our principles of academic integrity."[40]

**For a concise list of additional resources that may help guide you and your school leaders in developing a robust, well-thought-out AI policy for your district see: Steps to help address plagiarism concerns and avoid the risk of cognitive offloading.**

When students engage meaningfully in each stage of the writing process, from brainstorming, outlining, and drafting to revising and reflecting, they are less likely to resort to plagiarism, whether intentional or AI-assisted.

Digital platforms like Google Docs and other similar applications can support this approach through built-in tools such as version history, which allows teachers to see how a document has evolved over time. For even greater visibility, consider using browser add-ons like the free Draftback Chrome Extension, which replays a video of a document's writing history. These tools don't just serve as deterrents; they can also become an important part of the process and support formative feedback, metacognition, and writing skills development over time.[41]

**Pair this type of approach with intentional instructional strategies:**

- Have students submit artifacts from each phase of their writing (e.g., annotated sources, handwritten outlines, peer feedback, etc.).
- Create writing prompts that are personal, specific, or reflective, making them harder to answer meaningfully with AI alone.
- Hold brief writing conferences or ask students to explain and reflect on their thinking through oral or written metacognitive reflections.
- Above all, shift the classroom culture from policing to coaching, helping students understand not just what plagiarism is, but why authentic work matters in developing their own voice and learning.

### 3e. GenAI Copyright and Licensing Guidance for School Leaders

**Copyright/Licensing Unknowns:** Understanding copyright laws is an important element of guiding the use of genAI and other AI technologies in the classroom because this is new technology, and there are not yet clear boundaries regarding how AI tools can use copyrighted materials to learn from, nor who owns content generated by AI tools. According to the BBC

article, *AI firm Anthropic Agrees to Pay Authors $1.5bn to Settle Piracy Lawsuit*, in September 2025, Anthropic (owner of the Claude chatbot) agreed to pay $1.5 billion dollars to book authors after the judge ruled the company had illegally downloaded and stored millions of copyrighted books. As of the release of this guidance, according to a widely publicized letter from the US Copyright Office (Previous Correspondence ID: 1-5GB561K

Re: Zarya of the Dawn Registration # VAu001480196), AI-generated content cannot be copyrighted unless a human author contributes a *measurably significant amount of creative input*. As companies continue to develop licenses on their products, it is essential for educators to reflect on the implications of copyright/licensing unknowns. The U.S Copyright Office webpage on *Copyright and Artificial Intelligence* provides ongoing updates and revisions to copyright law and policy issues related to AI. Be sure to also:

- Review licensing types on the Creative Commons website and discuss copyright and licensing information with staff and students.
- Review the *US Copyright Office's New Artificial Intelligence Initiative*, which will continue to have the most up-to-date information on legal developments regarding copyright and generative AI. While not specific to education, as educators often use, curate, and share instructional materials through digital means, understanding copyright laws and how they impact the use of information developed through AI will be essential.

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